

DATA WAREHOUSING

- Data warehouse refers to a data repository that is maintained separately from an organization's operational databases
- The major task of online operational database systems is to perform online transaction and query processing
- They cover most of the day-to-day operations of an organization such as purchasing, inventory, manufacturing, banking, payroll, registration and accounting
- Data warehouse systems serve users in the role of data analysis and decision making
- They can organize and present data in various formats in order to accommodate the diverse needs of different users

Property	Database	Data Warehouse
Purpose	most useful for the small, atomic transactions	best suited for larger questions that require a higher level of analysis
Processing Method	Online Transaction Processing (OLTP)	Online Analytical Processing (OLAP)
Optimization	Deletes, inserts, updates large numbers of short online transactions quickly	Rapidly analyze massive volumes of data and provide different viewpoints for analysts
Data structure	Highly normalized data structure with many different tables containing no redundant data. Thus, data is more accurate	Denormalized data structure with few tables containing repeat data. Thus, data is potentially less accurate
Data analysis	Analysis is slow and painful due to the large number of table joins needed and the small time frame of data available	Analysis is fast and easy due to the small number of table joins needed and the extensive time frame of data available
Data timeline	Current, real-time data for one part of the business; historical queries impossible	Historical data for all parts of the business
Users	Can handle thousands of users at one time	Can only handle a smaller number

What Is a Data Warehouse?

- Data warehousing provides architectures and tools for business executives to systematically organize, understand, and use their data to make strategic decisions
- With competition mounting in every industry, data warehousing is the latest must-have marketing weapon—a way to retain customers by learning more about their needs
- Data warehouse systems allow for integration of a variety of application systems
- They support information processing by providing a solid platform of consolidated historic data for analysis

What Is a Data Warehouse?

- William H. Inmon, (leading architect in the construction of data warehouse systems) defines it as:

“A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management’s decision making process”

- Keywords—*subject-oriented, integrated, time-variant, and nonvolatile*—distinguish data warehouses from other data repository systems, such as relational database systems, transaction processing systems, and file systems

Subject-oriented:

- A data warehouse is organized around major subjects (e.g: customer, supplier, product, and sales).
- A data warehouse focuses on the modeling and analysis of data for decision makers (rather than concentrating on the day-to-day operations and transaction processing)
- Hence, data warehouses typically provide a simple and concise view of particular subject issues by excluding data that are not useful in the decision support process

Integrated:

- A data warehouse is usually constructed by integrating multiple heterogeneous sources, such as relational databases, flat files, and online transaction records
- Data cleaning and data integration techniques are applied to ensure consistency in naming conventions, attribute measures, and so on

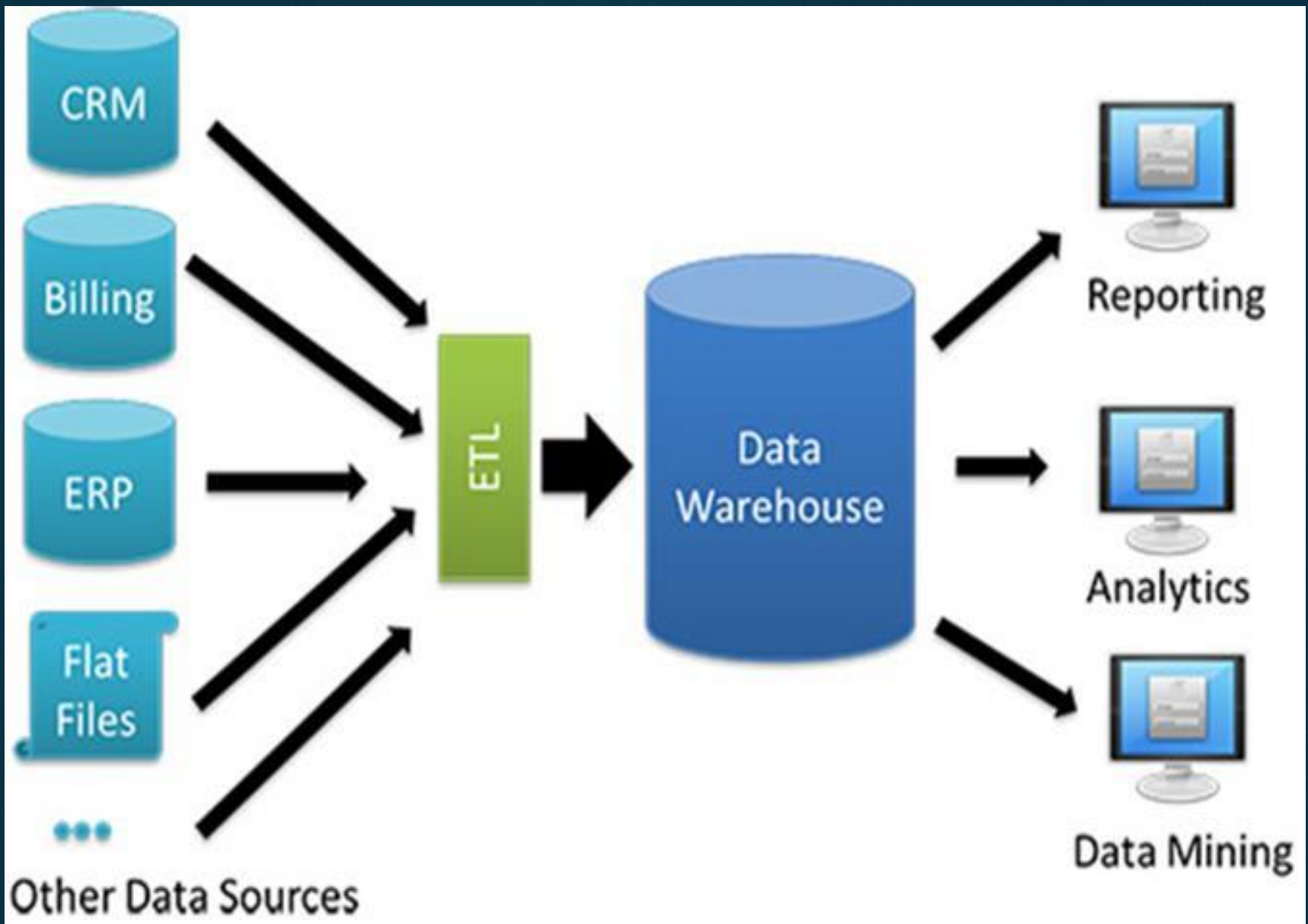
Time-variant:

- Data are stored to provide information from an historic perspective (e.g., the past 5–10 years)

Nonvolatile:

- A data warehouse is always a physically separate store of data transformed from the application data found in the operational environment
- Due to this separation, a data warehouse does not require transaction processing, recovery, and concurrency control mechanisms

- A data warehouse is a consistent data store that serves as a physical implementation of a decision support data model
- It stores the information that an enterprise needs to make strategic decisions
- A data warehouse is also often viewed as an architecture, constructed by integrating data from multiple heterogeneous sources to support structured and/or ad hoc queries, analytical reporting, and decision making.



- **Data warehousing** is the process of constructing and using data warehouses
- The **construction** of a data warehouse requires data cleaning, data integration, and data consolidation
- The **utilization** of a data warehouse often necessitates a collection of decision support technologies
- This allows “knowledge workers” (e.g., managers, analysts, and executives) to use the warehouse to quickly and conveniently obtain an overview of the data, and to make sound decisions based on information in the warehouse

How are organizations using the information from data warehouses?

Organizations use information to support business decision-making activities:

- Increasing customer focus, which includes the analysis of customer buying patterns
- Repositioning products and managing product portfolios by comparing the performance of sales by quarter, by year, and by geographic regions in order to fine-tune production strategies
- Analyzing operations and looking for sources of profit
- Managing customer relationships, making environmental corrections, and managing the cost of corporate assets

Differences between OLTP and OLAP

System orientation and Users:

- OLTP: *customer-oriented*; used for transaction and query processing by clerks, clients, and information technology professionals
- OLAP: *market-oriented*; used for data analysis by knowledge workers, including managers, executives, and analysts

Data contents:

- OLTP : manages current data that, typically, are too detailed to be easily used for decision making
- OLAP: manages large amounts of historic data, provides facilities for summarization and aggregation, and stores and manages information at different levels of granularity. These features make the data easier to use for informed decision making.

Database design:

- OLTP: usually adopts an entity-relationship (ER) data model and an application-oriented database design
- OLAP: typically adopts either a *star* or a *snowflake* model and a subject-oriented database design

Access patterns:

- OLTP: accesses consist mainly of short, atomic transactions. Such a system requires concurrency control and recovery mechanisms
- OLAP: accesses to OLAP systems are mostly read-only operations

View:

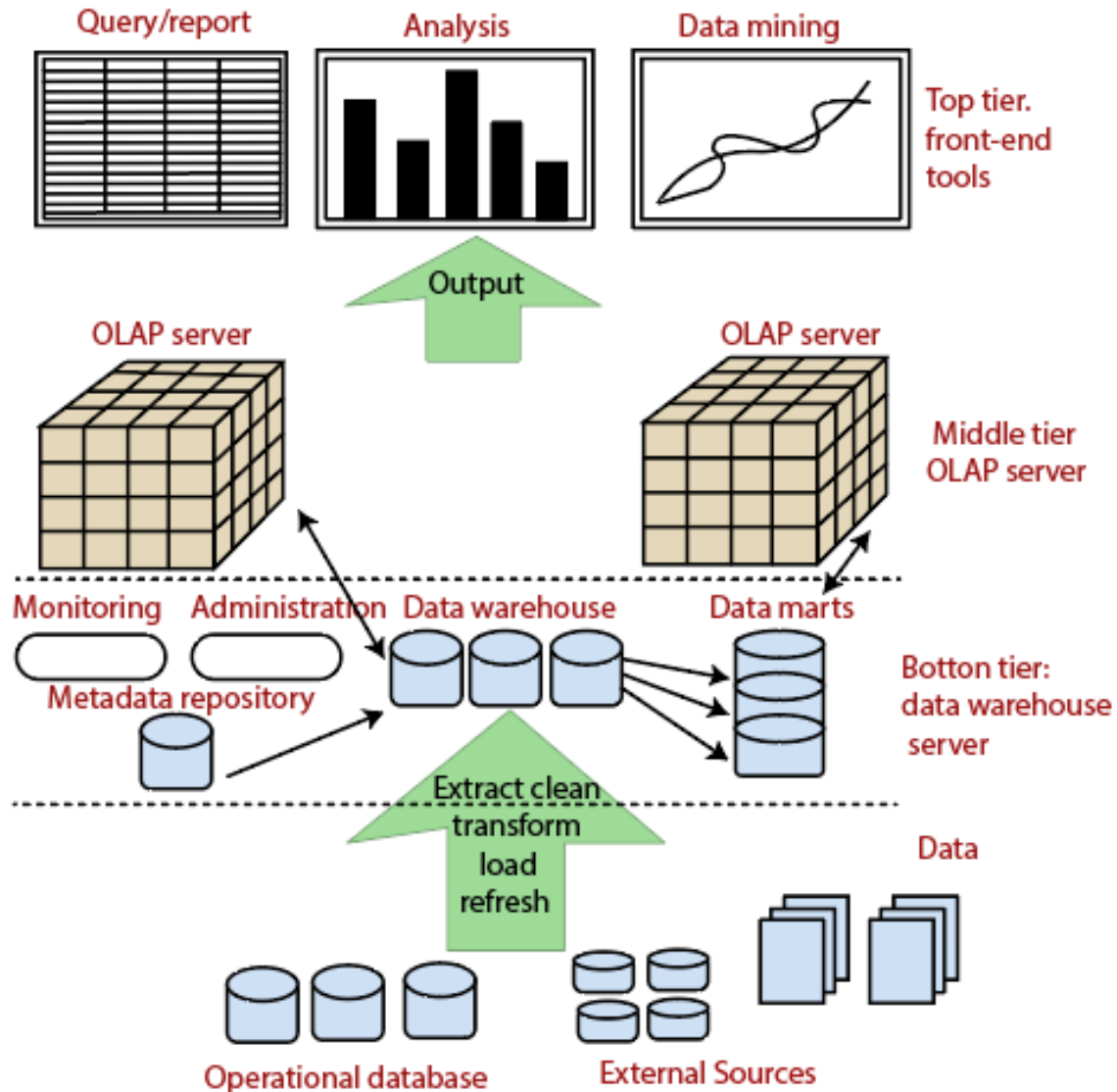
- OLTP: focuses mainly on the current data within an enterprise or department, without referring to historic data or data in different organizations
- OLAP: often spans multiple versions of a database schema, due to the evolutionary process of an organization. OLAP systems also deal with information that originates from different organizations, integrating information from many data stores

<i>Feature</i>	<i>OLTP</i>	<i>OLAP</i>
Characteristic	operational processing	informational processing
Orientation	transaction	analysis
User	clerk, DBA, database professional	knowledge worker (e.g., manager, executive, analyst)
Function	day-to-day operations	long-term informational requirements decision support
DB design	ER-based, application-oriented	star/snowflake, subject-oriented
Data	current, guaranteed up-to-date	historic, accuracy maintained over time
Summarization	primitive, highly detailed	summarized, consolidated
View	detailed, flat relational	summarized, multidimensional
Unit of work	short, simple transaction	complex query
Access	read/write	mostly read
Focus	data in	information out
Operations	index/hash on primary key	lots of scans
Number of records accessed	tens	millions
Number of users	thousands	hundreds
DB size	GB to high-order GB	$\geq$ TB
Priority	high performance, high availability	high flexibility, end-user autonomy
Metric	transaction throughput	query throughput, response time

Data Warehousing: A Multitiered Architecture

Data Warehouses usually have a three-level (tier) architecture that includes:

1. Bottom Tier (Data Warehouse Server)
2. Middle Tier (OLAP Server)
3. Top Tier (Front end Tools)



Three-Tier Data Warehouse Architecture

Bottom tier: Data warehouse Server

- Back-end tools and utilities are used to feed data into the bottom tier from operational databases or other external sources
- These tools and utilities perform data extraction, cleaning, and transformation as well as load and refresh functions to update the data warehouse
- The data are extracted using application program interfaces known as **Gateways**
 - A gateway is supported by the underlying DBMS
 - It allows client programs to generate SQL code to be executed at a server
 - Examples: ODBC (Open Database Connection), OLEDB (Object Linking and Embedding Database), JDBC (Java Database Connection)

- This tier also contains a metadata repository, which stores information about the data warehouse and its contents
- It includes the following parameters and information for the middle and the top-tier applications:
 - A description of the DW structure, including the warehouse schema, dimension, hierarchies, data mart locations, and contents, etc.
 - Operational metadata, which usually describes the currency level of the stored data, i.e., active, archived or purged, and warehouse monitoring information, i.e., usage statistics, error reports, audit, etc.
 - System performance data, which includes indices, used to improve data access and retrieval performance.
 - Information about the mapping from operational databases, which provides source RDBMSs and their contents, cleaning and transformation rules, etc.

Middle tier: OLAP server

- Data Warehouse can have more than one OLAP server, and it can have more than one type of OLAP server model as well, which depends on the volume of the data to be processed and the type of data held in the bottom tier
- OLAP server is typically implemented using either:
- **Relational OLAP(ROLAP) model**
 - An extended relational DBMS that maps operations on multidimensional data to standard relational operations
- **Multidimensional OLAP (MOLAP) model**
 - A special-purpose server that directly implements multidimensional data and operations

Top tier: front-end client layer

- Includes Tools and APIs that connects and get data out from the data warehouse and present the same to the client
- Contains query and reporting tools, analysis tools, data mining tools (e.g., trend analysis, prediction etc.)
- The type of tool depends purely on the form of outcome expected

Data Warehouse Models

From the architecture point of view, there are three data warehouse models:

- Enterprise warehouse
- Data mart
- Virtual warehouse

Enterprise warehouse

- It collects all the information about subjects spanning the entire organization
- It provides corporate-wide data integration, usually from one or more operational systems or external information providers
- It is cross-functional in scope
- It typically contains detailed data as well as summarized data, and can range in size from a few gigabytes to hundreds of gigabytes, terabytes, or beyond
- An enterprise data warehouse may be implemented on traditional mainframes, computer superservers, or parallel architecture platforms
- It requires extensive business modeling and may take years to design and build

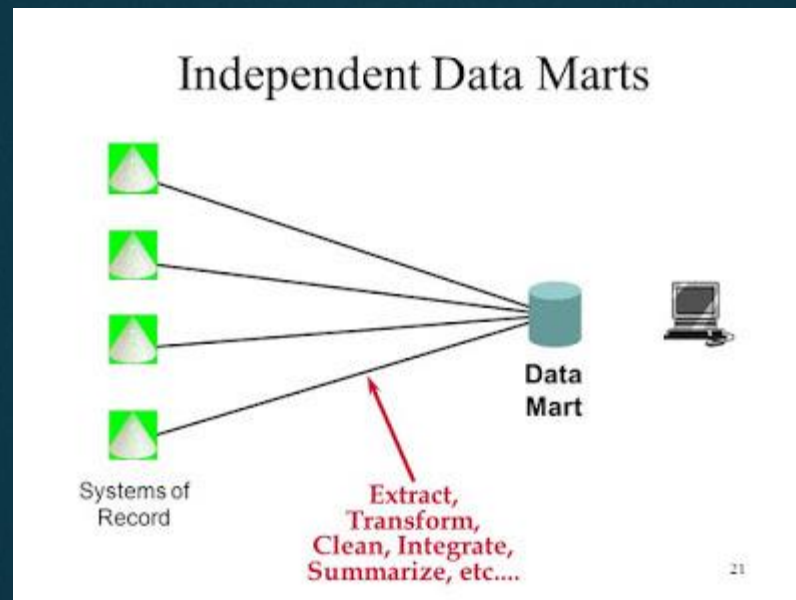


Data mart

- A data mart contains a subset of corporate-wide data that is of value to a specific group of users
- The scope is confined to specific selected subjects
- Example: A marketing data mart may confine its subjects to customer, item, and sales.
- The data contained in data marts tend to be summarized
- Data marts are usually implemented on low-cost departmental servers that are Unix/Linux or Windows based
- The implementation cycle of a data mart is more likely to be measured in weeks rather than months or years

- Independent data marts:

- Sourced from data captured from one or more operational systems or external information providers, or from data generated locally within a particular department or geographic area



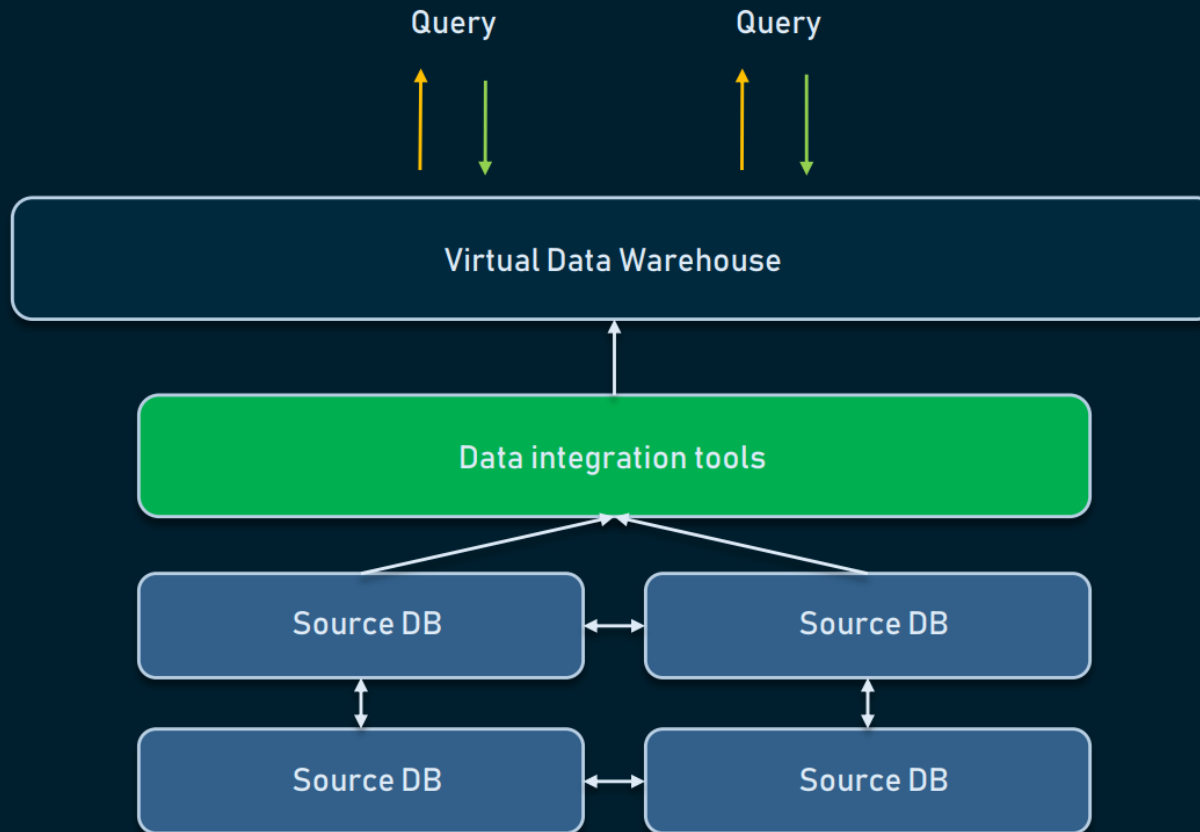
- **Dependent data marts:**
 - Sourced directly from enterprise data warehouses



Virtual Warehouse

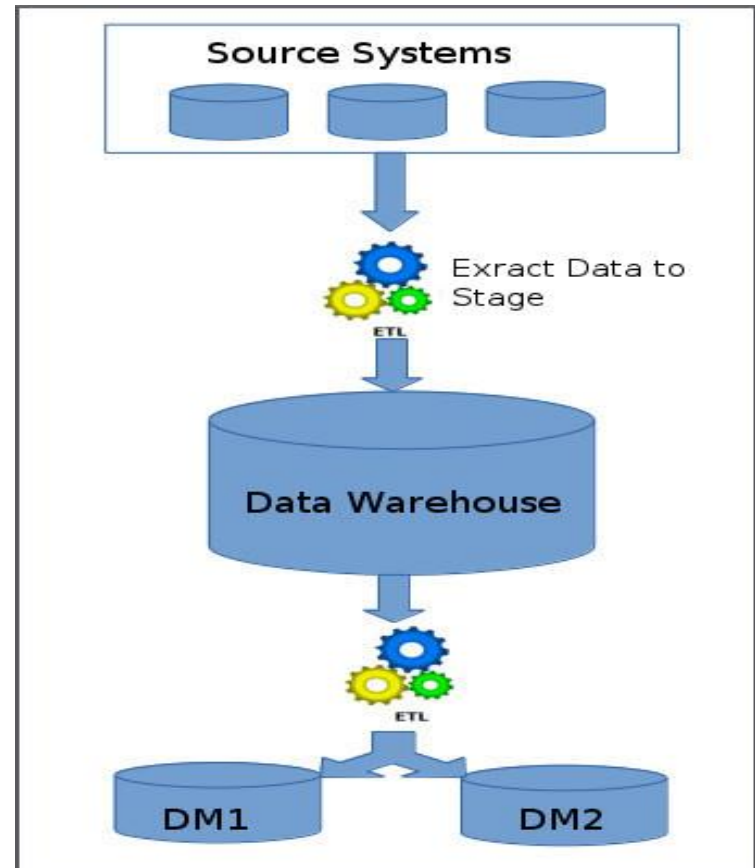
- A virtual warehouse is a set of views over operational databases
- For efficient query processing, only some of the possible summary views may be materialized
- It uses middleware to build connections to different data sources
- Set of separate databases, which can be queried together, so a user can effectively access all the data as if it was stored in one data warehouse- *A layer of virtuality is created*
- They can be fast as they allow users to filter the most important pieces of data from different legacy applications
- A virtual warehouse is easy to build but requires excess capacity on operational database servers

VIRTUAL DATA WAREHOUSE



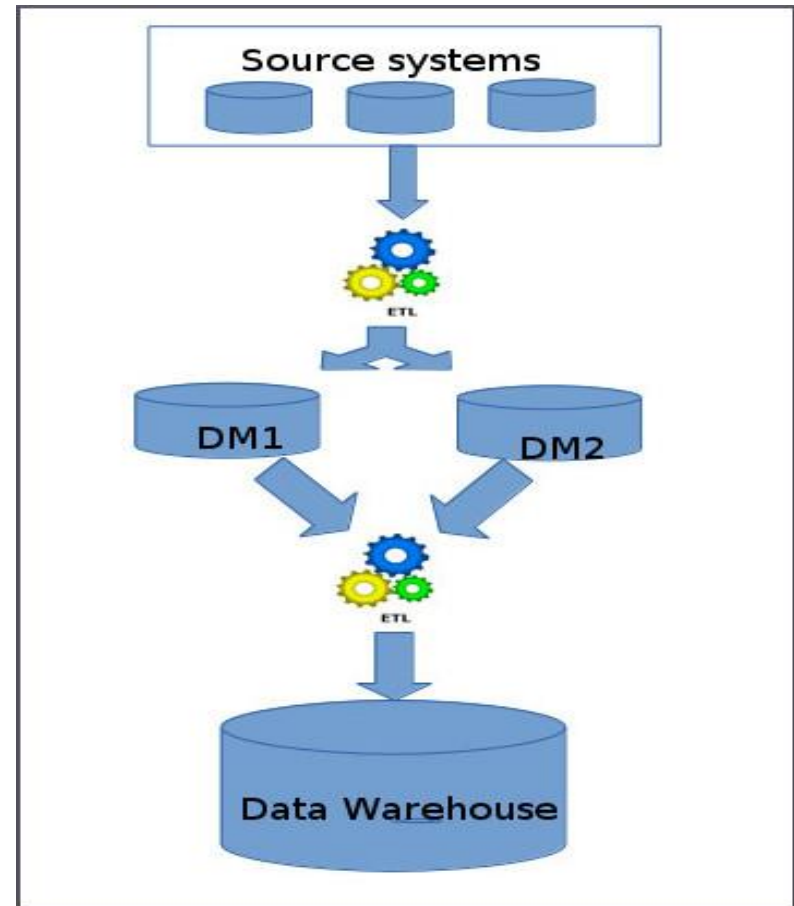
Approaches to the DW development

- ▶ **Top-down Approach(Bill Inmon)**
 - Data warehouse is designed first and then data mart are built on top of data warehouse
 - Serves as a systematic solution
 - Minimizes integration problems
 - Expensive
 - Takes a long time to develop
 - Lacks flexibility due to the difficulty in achieving consistency and consensus for a common data model for the entire organization



► Bottom-up Data Approach(Ralph Kimball)

- Data marts are first created to provide the reporting and analytics capability for specific business process, later with these data marts enterprise data warehouse is created
- Development and deployment of independent data marts provides flexibility, low cost, and rapid return of investment
- It can lead to problems when integrating various disparate data marts into a consistent enterprise data warehouse



(Kimball Approach)

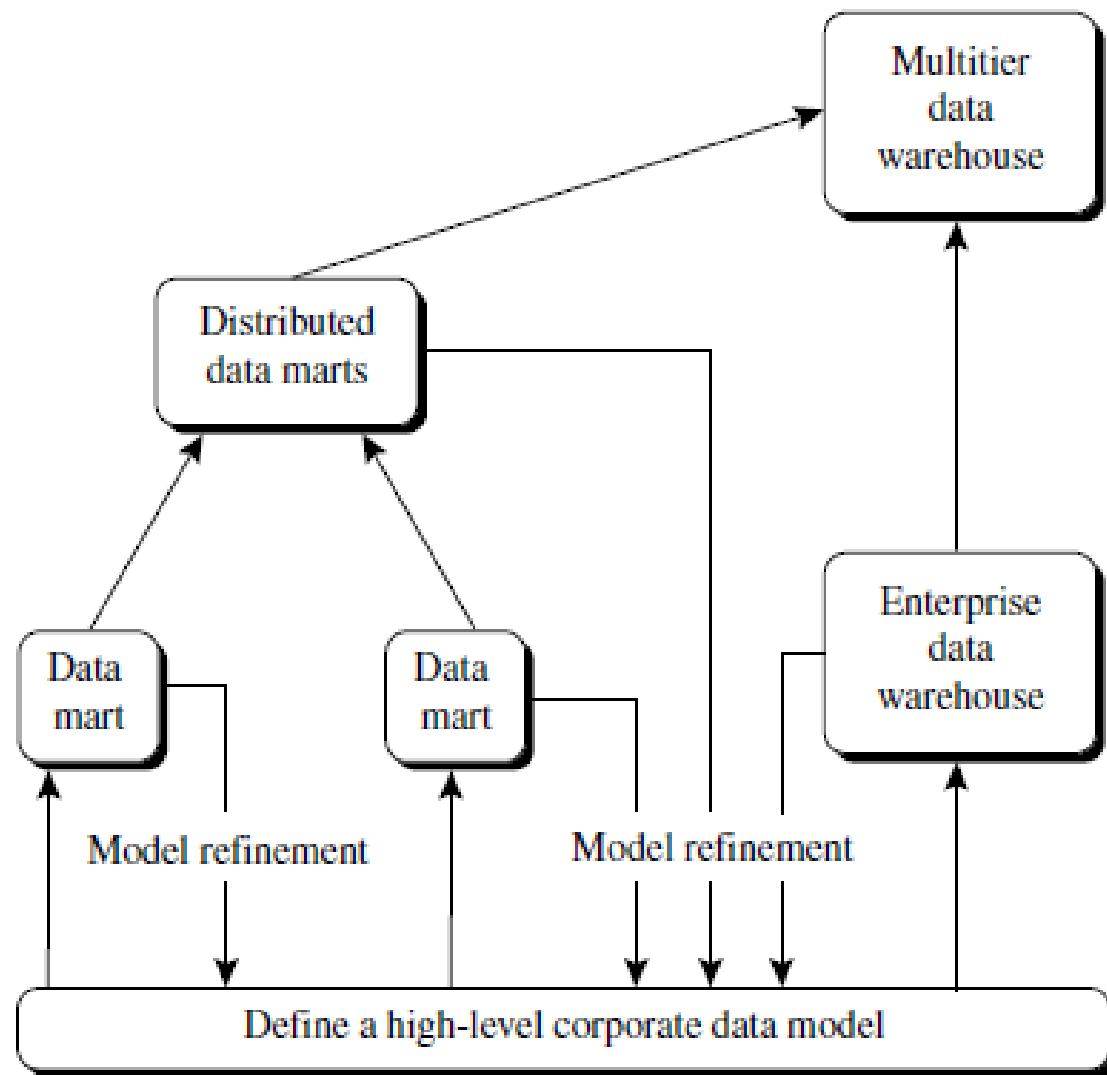
(Inmon Approach)

	Functional Data Mart	Enterprise Data Warehouse
Scope	One subject or functional area	Complete enterprise data needs
Value	Functional area reporting and insights	Deeper insights connecting multiple functional areas
Target organization	Decentralized management	Centralized management
Time	Low to medium	High
Cost	Low	High
Size	Small to medium	Medium to large
Approach	Bottom up	Top down
Complexity	Low (fewer data transformations)	High (data standardization)
Technology	Smaller scale servers and databases	Industrial strength

Recommended method for the development of DW

Implement the warehouse in an incremental and evolutionary manner

- A high-level corporate data model is defined within a reasonably short period
 - provides a corporate-wide, consistent, integrated view of data among different subjects and potential usages
 - need to be refined in the further development of enterprise data warehouses and departmental data marts
 - greatly reduce future integration problems
- Independent data marts can be implemented in parallel with EDW based on the corporate data model
- Distributed data marts can be constructed to integrate different data marts via hub servers
- A multitier data warehouse is constructed
 - where the enterprise warehouse is the sole custodian of all warehouse data, which is then distributed to the various dependent data marts

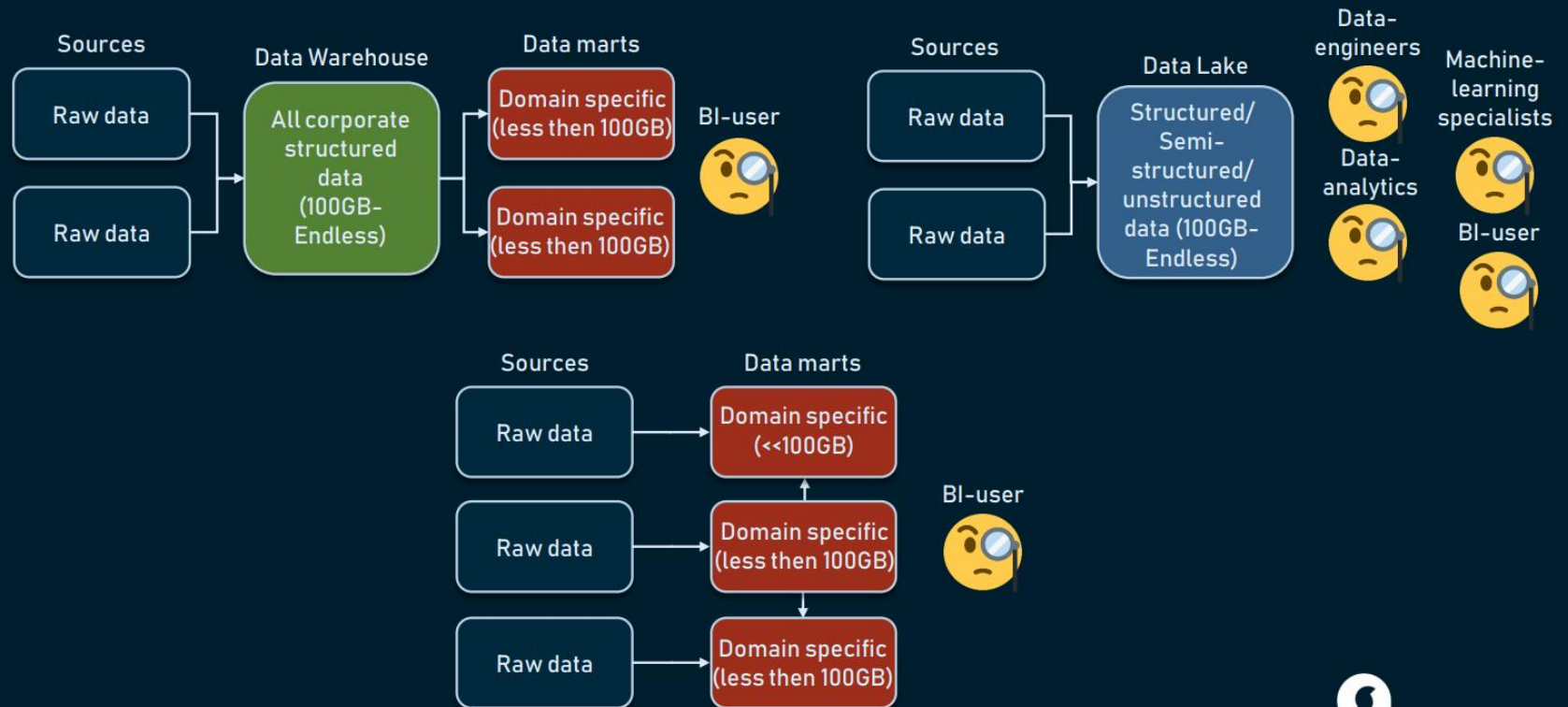


Data Lakes

- A data lake is a storage repository that holds a vast amount of raw data in its native format until it is needed
- It is usually a single store of data including raw copies of source system data, sensor data, social data etc., and transformed data used for tasks such as reporting, visualization, advanced analytics and machine learning.
- A data lake can include:
 - structured data from relational databases (rows and columns),
 - semi-structured data (CSV, logs, XML, JSON),
 - unstructured data (emails, documents, PDFs) and
 - binary data (images, audio, video)
- A data lake can be established "on premises" or "in the cloud"

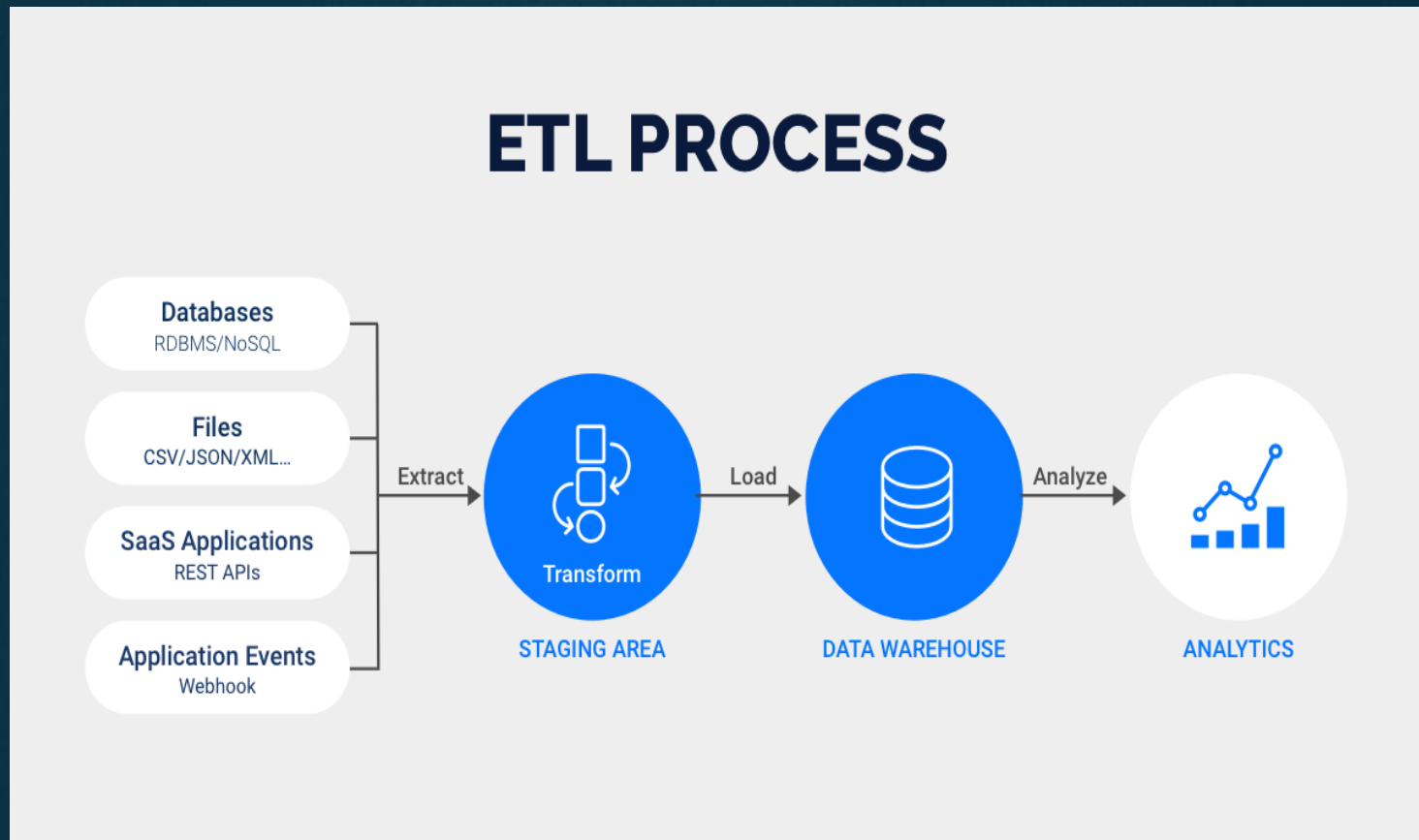
Parameters	Data Lake	Data Warehouse
Data Structure	Data is raw and all types—structured, semi-structured, or unstructured—is captured in its original form.	Data is processed and only structured information is captured and organized in schemas.
Users	Ideal for users who carry out deep analysis such as data scientists and need advanced analytical tools.	Ideal for operational users such as business professionals and moguls since the data is structured and easy to use.
Storage Costs	Storing data is relatively inexpensive.	Storing data is time-consuming and costly.
Accessibility	Updates can be made quickly thus making it highly accessible	Costly to make changes, thereby quite complicated
Position of Schema	Schema is defined after data is stored, thus making it highly agile.	Schema is defined before data is stored, thus offering performance and security.
Data Processing	Uses ELT (Extract Load Transform) process.	Uses ETL (Extract Transform Load) process.

DATA WAREHOUSE VS DATA LAKE VS DATA MART



Extraction, Transformation, and Loading

- ETL is a process that extracts the data from different source systems, then transforms the and finally loads the data into the Data Warehouse system



Extraction

- Process of extracting the data from the source system(s)
- Extracting data correctly sets the stage for the success of subsequent processes
- Most data-warehousing projects combine data from different source systems- relational databases, XML, JSON and flat files
- Data warehouse needs to integrate systems that have different
- DBMS, Hardware, Operating Systems and Communication Protocols
- Sources could include legacy applications like Mainframes, customized applications, Point of contact devices like ATM, Call switches, text files, spreadsheets, ERP, data from vendors, partners amongst others

Examples for validations done during Extraction:

- Make sure that no spam/unwanted data loaded
- Data type check
- Remove all types of duplicate/fragmented data

Transformation

- In the data transformation stage, a series of rules or functions are applied to the extracted data in order to prepare it for loading into the end target
- Data extracted from source server is raw and not usable in its original form. Therefore it needs to be cleansed, mapped and transformed
- This step adds value and changes data such that insightful BI reports can be generated
- Transformations if any are done in staging area so that performance of source system is not degraded
- Also, if corrupted data is copied directly from the source into Data warehouse database, rollback will be a challenge.
- Staging area gives an opportunity to validate extracted data before it moves into the Data warehouse

Examples for transformations:

- Selecting only certain columns to load:
- Translating coded value(ssource system codes male as "1" and female as "2", but the warehouse codes male as "M" and female as "F")
- Encoding free-form values: (e.g., mapping "Male" to "M")
- Deriving a new calculated value: (e.g., $\text{sale_amount} = \text{qty} * \text{unit_price}$)
- Sorting or ordering the data based on a list of columns to improve search performance
- Aggregating
- Splitting a column into multiple columns
- Applying any form of data validation

Loading

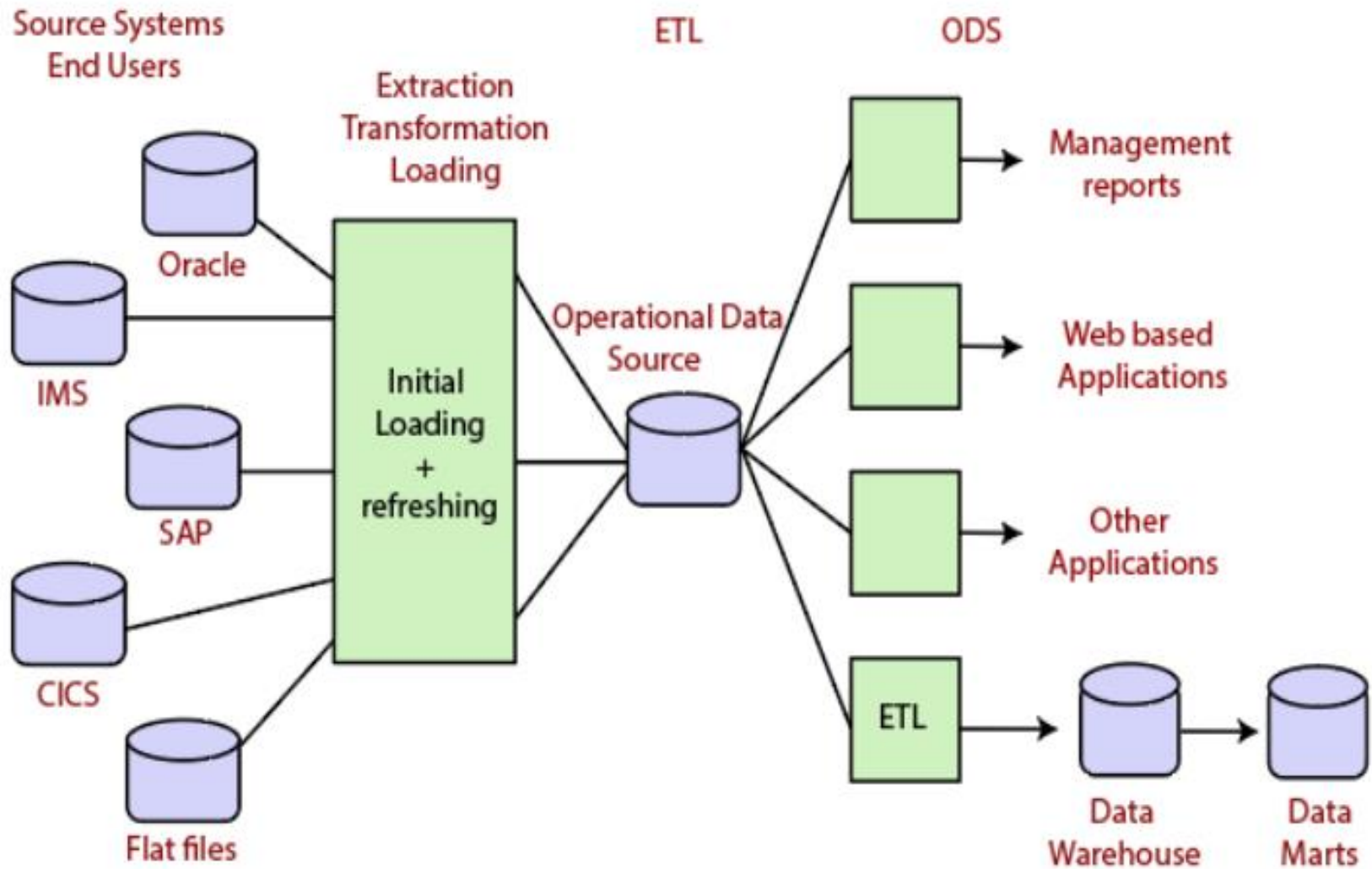
- Process of Loading data into the target datawarehouse database
- In a typical Data warehouse, huge volume of data needs to be loaded in a relatively short period. Hence, load process should be optimized for performance.
- In case of load failure, recover mechanisms should be configured to restart from the point of failure without data integrity loss
- Updating extracted data is frequently done on a daily, weekly, or monthly basis
- Some data warehouses may overwrite existing information with cumulative information
- Other data warehouses (or even other parts of the same data warehouse) may add new data in a historical form at regular intervals — for example, hourly

- **Data cleaning:** detects errors in the data and rectifies them when possible.
- **Refresh:** propagates the updates from the data sources to the warehouse
- Examples for ETL tools: Xplenty, Talend, Stitch, Informatica PowerCenter, Oracle Data Integrator, Skyvia etc

OPERATIONAL DATA STORES(ODS)

- A subject-oriented, integrated, volatile, current valued data store, containing only corporate detailed data
- **Volatile:** data in the ODS changes frequently as new information refreshes the ODS
- **Current valued:** an ODS is up-to-date and reflects the current status of the information
- Since the OLTP systems data is changing all the time, data from underlying sources refresh the ODS as regularly and frequently as possible
- **Detailed:** ODS is detailed enough to serve the needs of the operational management staff in the enterprise

Operational Data Stores	Data Warehouse
ODS means for operational reporting and supports current or near real-time reporting requirements.	A data warehouse is intended for historical and trend analysis, usually reporting on a large volume of data.
An ODS consist of only a short window of data .	A data warehouse includes the entire history of data .
It is typically detailed data only.	It contains summarized and detailed data.
It is used for detailed decision making and operational reporting.	It is used for long term decision making and management reporting.
It is used at the operational level.	It is used at the managerial level.
It is updated often as the transactions system generates new data.	It is usually updated in batch processing mode on a set schedule.



Operational Data Store Structure